

KHUSHBOO PANDEY

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EDUCATION

University of Massachusetts Amherst

Master of Science in Computer Science

Amherst, MA

Expected Graduation: May 2027

- Relevant Coursework: Distributed Operating Systems, Software Engineering, Machine Learning

Bikaner Technical University

Bachelor of Technology in Electronics & Communication Engineering

Bikaner, India

Graduated: 2018

- Certifications: Datadog Certified (Cloud Monitoring, APM, Logging & Security Alerts); Terraform Basics – KodeKloud (Infrastructure as Code, AWS Provisioning & Workflows); Artificial Intelligence Foundations – Accenture

TECHNICAL SKILLS

Cloud Platforms: AWS (EC2, S3, Lambda, VPC, IAM, EKS, RDS, CloudFront, CloudWatch, Secrets Manager, KMS), Google Cloud Platform

Infrastructure as Code: Terraform, AWS CDK, AWS CloudFormation, Ansible

Containers & Orchestration: Docker, Kubernetes (EKS), Helm, Nginx

CI/CD & Automation: GitHub Actions, Jenkins, Harness, Azure DevOps, Git

Security: IAM least-privilege design, AWS Security Hub, SonarQube, KMS encryption, secrets management

Monitoring & Observability: Datadog, Prometheus, Grafana, Amazon CloudWatch, AWS Cost Explorer

Programming & Data: Python, Bash/Shell, JavaScript/Node.js, C/C++, SQL, MongoDB, PostgreSQL, Elasticsearch

PROFESSIONAL EXPERIENCE

Cloud Engineer, Volansys Technologies (Client: Arlo Technologies)

Apr 2023 – Jan 2026

- Gathered infrastructure requirements from engineering, security, and product stakeholders and translated them into Confluence architecture standards, implementation guides, and security posture documentation used org-wide.
- Built and maintained Infrastructure-as-Code modules in Terraform and CloudFormation to provision AWS networking, compute, and serverless resources (VPC, EC2, EKS, Lambda) for enterprise video-streaming and surveillance workloads, evaluating each build against AWS Well-Architected Framework best practices across all six pillars.
- Kept five production services at a 99.9% uptime SLA by setting up continuous monitoring and alerting through Datadog, Grafana, and Amazon CloudWatch, and proactively investigating reliability bottlenecks.
- Strengthened the security posture of client environments by rolling out centralized secrets management with AWS Secrets Manager, KMS-based encryption for S3 and RDS, static code scanning with SonarQube in the CI/CD pipeline, and real-time CloudWatch security alerts routed to Slack and PagerDuty.
- Cut mean time to detect security issues by 40% by codifying alert routing and monitoring dashboards as Terraform-managed configuration, ensuring consistent monitoring setup across staging and production.
- Acted as primary War Room responder for high-severity production incidents, driving root-cause analysis, coordinating fixes across teams, and writing blameless postmortems that led to permanent corrective changes.
- Lowered content-delivery latency by 30% by tuning AWS CloudFront edge-caching configurations for high-traffic streaming workloads.

Cloud Engineer, Volansys Technologies (Client: Rheem Manufacturing)

Jun 2022 – Apr 2023

- Designed and delivered cloud solutions on AWS and GCP for an enterprise manufacturing client using Terraform and AWS CDK, cutting monthly infrastructure spend by roughly \$12,000 while improving scalability.
- Audited the client's cloud environment against AWS Well-Architected Framework standards and closed a cost-efficiency gap by right-sizing over-provisioned compute and storage resources.
- Stood up Datadog and Grafana alerting tailored to IoT-based heating and cooling control platforms, shrinking incident resolution time and supporting more reliable release cycles.

Cloud Engineer, Promaestro Techsource (Client: Techmentor IoT Solutions)

Nov 2019 – May 2022

- Modularized Terraform scripts for core AWS infrastructure (EC2, S3, RDS), encoding security and reliability standards into reusable modules and speeding up deployments by 50%.
- Built and maintained Jenkins CI/CD pipelines with automated quality gates and rollback triggers, enabling reliable, traceable releases for high-traffic production IoT microservices.

- Monitored application performance and troubleshoot infrastructure issues using Datadog, cutting incident response time by 90% and keeping delivery consistent across high-volume production environments.

PROJECTS

AI-Driven AWS Cost Analyzer

Python, AWS APIs, LLM Integration, Terraform – github.com/khushboopandey6/ai-cloud-cost-analyzer

- Built a FinOps tool that pulls AWS Cost Explorer and CloudWatch data through a serverless ingestion pipeline to surface idle EC2 instances and unattached EBS volumes.
- Layered an LLM on top of the usage data to generate cost-reduction recommendations aligned with the AWS Well-Architected Cost Optimization pillar, surfacing up to 25% in potential savings.

ReLoop – Sustainability & Re-Commerce Platform

Node.js, Express, MongoDB, AWS S3, Docker – github.com/khushboopandey6/ReLoop

- Built the RESTful backend for a re-commerce marketplace application, handling listings, transactions, and user workflows.
- Wired up AWS S3 through multer-s3 middleware for media storage, keeping binary assets out of the application database and served via secure S3 object URLs.
- Containerized the application's microservices with Docker for consistent behavior across local development and cloud deployment.